

A Memoir on the Transformation of Elliptic Functions

(Continuation of paper 578 — pp. 118–142 of Vol. IX)

[p. 118] Tables for the cases $n = 7$ and $n = 11$ (*continued*)

[Table: the second equation-system for the septic case $n = 7$. The rows are labelled by powers Ω^k ($k = 0, \dots, 8$) and the columns by $k^9, k^8, k^7, k^6, k^5, k^4, k, 1$. The non-zero entries are arranged so that the equation reads, on summation by rows,]

$$\begin{aligned}\Omega^0 &: +1, \\ \Omega^1 &: -64k^8 + 56k^6, \\ \Omega^2 &: -112k^7 + 140k^5, \\ \Omega^3 &: -112k^6 + 56k^4, \\ \Omega^4 &: +70k^5, \\ \Omega^5 &: +56k^4 - 112k^2, \\ \Omega^6 &: +140k^3 - 112k, \\ \Omega^7 &: +56k^2 - 64, \\ \Omega^8 &: +1,\end{aligned}$$

the column-sums of which (read in the order $k^9, k^8, k^7, k^6, k^5, k^4, k, 1$) are

$$-64, -112, 0, +352, 0, -112, -64, = 71 \cdot 11;$$

that is, $\Omega = 1$, we have

$$-16(k-1)^2(4k^2 + 3k + 1)(k^2 + 3k + 4) = 0.$$

[Table: the equation-system for $n = 11$, endecadic transformation. Rows are labelled $\Omega^0, \dots, \Omega^{12}$ and columns by $k^{10}, k^9, k^8, k^7, k^6, k^5, k^4, k^3, k^2, k, 1$. The non-zero entries, row by row, are:]

$$\begin{aligned}\Omega^0 &: +1, \\ \Omega^1 &: -1024k^{10} + 1408k^9 - 396k^8 - 12, \\ \Omega^2 &: -5632k^9 + 4400k^8 + 1298k^7 + 66, \\ \Omega^3 &: -16192k^8 + 16368k^7 - 396k^5 - 220, \\ \Omega^4 &: -18656k^7 + 19151k^6 + 495, \\ \Omega^5 &: -16016k^6 - 1144k^5 + 16368k^4 - 792, \\ \Omega^6 &: +4400k^5 - 7876k^4 + 4400k^3 + 924, \\ \Omega^7 &: +16368k^4 - 1144k^3 - 16016k^2 - 792, \\ \Omega^8 &: +19151k^3 - 18656k^2 + 495, \\ \Omega^9 &: -396k^4 + 16368k^2 - 16192k - 220, \\ \Omega^{10} &: +1298k^3 + 4400k - 5632 + 66, \\ \Omega^{11} &: -396k^2 + 1408 - 1024 - 12, \\ \Omega^{12} &: +1,\end{aligned}$$

the column-sums of which (read across the columns $k^{10}, k^9, \dots, k, 1$) are

$$\begin{aligned}&-1024, -32208, -18656, -1936, -7876, -1936, -18656, -32208, -1024, \\ &+1408, +8800, +32736, +40900, +32736, +8800, +1408, \\ &-5632, -30800, -9856, +30800, +33024, +30800, -9856, -30800, -5632, \\ &-1024 + 94624.\end{aligned}$$

[p. 119] Equation-systems for the cases $n = 3, 5, 7, 9, 11$. Art. Nos. 8 to 10.

8. $n = 3$, cubic transformation. $k = u^4$, $\Omega = \frac{u^9}{v}$ (here and in the other cases).

$P = \alpha$, $Q = \beta$. The condition here is

$$k^2 \alpha^2 \alpha^2 + (2\alpha\beta + \beta^2)^2 = \Omega[(\alpha^2 + 2\alpha\beta)^2 + \beta^2 \alpha^2],$$

and the system of equations thus is

$$k\alpha^2 = \Omega\beta^2, \quad 2\alpha\beta + \beta^2 = \Omega k(\alpha^2 + 2\alpha\beta),$$

and similarly in the other cases; for these it will be enough to write down the equation-systems.

$n = 5$, quintic transformation.

$$P = \alpha + \gamma x^2, \quad Q = \beta.$$

$$k^2 \alpha^2 = \Omega \gamma^2,$$

$$2\alpha\gamma + 2\alpha\beta + \beta^2 = \Omega(2\beta\gamma + 2\beta\gamma + \beta^2),$$

$$\gamma^2 + 2\beta\gamma = \Omega k^2(\alpha^2 + 2\alpha\beta).$$

$n = 7$, septic transformation.

$$P = \alpha + \gamma x^2, \quad Q = \beta + \delta x^2.$$

$$k^2 \alpha^2 = \Omega \delta^2,$$

$$k(2\alpha\gamma + 2\alpha\beta + \beta^2) = \Omega(\gamma^2 + 2\gamma\delta + 2\beta\delta),$$

$$\gamma^2 + 2\beta\gamma + 2\alpha\delta + 2\beta\delta = \Omega k(2\alpha\gamma + 2\beta\gamma + 2\alpha\delta + \beta^2),$$

$$\delta^2 + 2\gamma\delta = \Omega k^2(\alpha^2 + 2\alpha\beta).$$

$n = 9$, enneadic transformation.

$$P = \alpha + \gamma x^2 + \epsilon x^4, \quad Q = \beta + \delta x^2.$$

$$k^2 \alpha^2 = \Omega \epsilon^2,$$

$$k(2\alpha\gamma + 2\alpha\beta + \beta^2) = \Omega(2\gamma\epsilon + 2\alpha\delta + \delta^2),$$

$$2\alpha\epsilon + \gamma^2 + 2\alpha\delta + 2\gamma\beta + 2\beta\delta = \Omega(2\alpha\epsilon + \gamma^2 + 2\gamma\delta + 2\delta\beta + 2\beta\delta),$$

$$2\gamma\epsilon + 2\gamma\delta + 2\epsilon\beta + \delta^2 = \Omega k(2\beta\gamma + 2\alpha\delta + 2\gamma\beta + \beta^2),$$

$$\epsilon^2 + 2\delta\epsilon = \Omega k^2(\alpha^2 + 2\alpha\beta).$$

$n = 11$, endecadic transformation.

$$P = \alpha + \gamma x^2 + \epsilon x^4, \quad Q = \beta + \delta x^2 + \zeta x^4.$$

$$k^2 \alpha^2 = \Omega \zeta^2,$$

$$k(2\alpha\gamma + 2\alpha\beta + \beta^2) = \Omega(\epsilon^2 + 2\epsilon\zeta + 2\delta\zeta),$$

$$k(2\alpha\epsilon + \gamma^2 + 2\alpha\delta + 2\gamma\beta + 2\beta\delta) = \Omega(2\gamma\epsilon + 2\gamma\zeta + 2\epsilon\delta + 2\beta\zeta + \delta^2),$$

$$2\gamma\epsilon + 2\alpha\zeta + 2\gamma\delta + 2\epsilon\beta + 2\beta\zeta + \delta^2 = \Omega k(2\alpha\epsilon + \gamma^2 + 2\alpha\zeta + 2\gamma\delta + 2\epsilon\beta + 2\beta\delta),$$

$$\epsilon^2 + 2\gamma\zeta + 2\epsilon\delta + 2\delta\zeta = \Omega k^2(2\alpha\gamma + 2\alpha\delta + 2\gamma\beta + \beta^2),$$

$$2\epsilon\zeta + \zeta^2 = \Omega k^2(\alpha^2 + 2\alpha\beta).$$

And so on.

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9. It will be noticed that if the coefficients of $P + Qx$ taken in order are

$$\alpha, \beta, \gamma, \dots, \sigma, \tau,$$

then in every case the first and last equations are

$$k\alpha^2 = \Omega\tau^2, \quad \tau^2 + 2\rho\sigma\tau = \Omega^{\frac{1}{2}(n-1)}k^{\frac{1}{2}(n-1)}(\alpha^2 + 2\alpha\beta).$$

Putting in the first of these $k = u^4$, $\Omega = \frac{u^n}{v}$, the equation becomes

$$\frac{u^4\alpha^2}{v} = \frac{u^n}{v}\tau^2,$$

where each side is a perfect square; and in extracting the square root we may without loss of generality take the roots positive, and write $\tau = u^{\frac{1}{2}(n-4)}\alpha = v\alpha$.

This speciality, although it renders it proper to employ ultimately u, v in place of k, Ω , produces really no depression of order (viz. the Ωk -form of the modular equation is found to be of the same order in Ω that the standard or uv -form is in v), and is in another point of view a disadvantage, as destroying the uniformity of the several equations: in the discussion of order I consequently retain Ω, k . Ultimately these are to be replaced by u, v ; the change in the equation-systems is so easily made that it is not necessary here to write them down in the new form in u, v .

10. The case $\alpha = 0$ has to be considered in the discussion of order, but we have thus only solutions which are to be rejected; in the proper solutions α is not $= 0$, and it may therefore for convenience be taken to be $= 1$. We have then $\sigma = u^{n-4} \div v$.

The last equation becomes therefore

$$\tau^2 + 2\rho\sigma\tau = \Omega^{\frac{1}{2}(n-1)}k^{\frac{1}{2}(n-1)}(1 + 2\beta),$$

or recollecting that β is connected with the multiplier M by the relation $\frac{1}{M} = 1 + 2\beta$, that is,

$$2\beta = \frac{1}{M} - 1,$$

and substituting for $1 + 2\beta$ its value, the equation becomes

$$2\rho = u^{4(n-1)}\left(\frac{1}{M} - \frac{u^n}{v^n}\right);$$

that is, the first and the last coefficients are 1, $\frac{u^{n-4}}{v}$, and the second and the penultimate coefficients are each expressed in terms of v, M . The cases $n = 3$, $n = 5$ are so far peculiar, that the only coefficients are α, β , or α, β, γ ; in the next case $n = 7$, the only coefficients are $\alpha, \beta, \gamma, \delta$, and we have in this case all the coefficients expressed as above.

[p. 121] The Ωk -form — Order of the Systems. Art. Nos. 11 to 22.

11. In the general case, n an odd number, we have Ω , and $\frac{1}{2}(n+1)$ coefficients connected by a system of $\frac{1}{2}(n+1)$ equations of the form

$$\Omega = \frac{U}{U'} = \frac{V}{V'} = \dots,$$

where $U, V, \dots, U', V', \dots$ are given quadric functions of the coefficients. Omitting the $\Omega (=)$, there remains a system of $\frac{1}{2}(n-1)$ equations of the form $\frac{U}{U'} = \frac{V}{V'} = \dots$, or say

$$\left| \begin{array}{ccc} U, & V, & W, \dots \\ U', & V', & W', \dots \end{array} \right| = 0,$$

which determine the ratios $\alpha : \beta : \gamma : \dots$ of the coefficients; and to each set of ratios there corresponds a single value of Ω . The order of the system, or number of sets of ratios, is $= \frac{1}{2}(n+1) \cdot 2^{\frac{1}{2}(n-1)}$, $= \frac{1}{2}(n+1) \cdot 2^{\frac{1}{2}(n-1)}$; and this is consequently the number of values of Ω , or the order of the equation for the determination of Ω ; viz. but for reduction, the order in Ω of the Ωk -modular equation would be $= (n+1) \cdot 2^{\frac{1}{2}(n-1)}$. In the case $n = 3$, this is 4, which is right, but for any larger value of n the order is far too high; in fact, assuming (as the case is) that the order is equal to the order in v of the uv -form, the order should for a prime value of n be $= n+1$, and for a composite value not containing any square factor be $=$ the sum of the divisors of n . I do not attempt a general investigation, but confine myself to showing in what manner the reductions arise.

12. I will first consider the cubic transformation; here, writing for convenience $\frac{\alpha}{\beta} = \theta$, the equations give

$$\frac{k\theta^2}{2\theta+1} = \frac{1}{k(\theta^2+2\theta)}, \quad \text{that is,} \quad k^2\theta^2(\theta+2) - (2\theta+1) = 0,$$

and

$$k\theta^2 = \Omega.$$

The equation in θ gives $(k^2\theta^2 - 1)^2 - 4\theta(k^2\theta^2 - 1) = 0$, and we have thence

$$k(\Omega^2 - 1)^2 = 4\Omega(k\Omega - 1)^2,$$

that is, $k(\Omega^2 - 1)^2 - 4\Omega(k\Omega - 1)^2 = 0$,

$$k\Omega^4 - 4k\Omega^3 + 6k\Omega^2 - 4\Omega + k = 0,$$

the modular equation; and then $-1 + k^2\theta^2 \mid 2\theta(k^2\theta^2 - 1) = 0$, that is, $\beta = 1 + 2\theta(k\Omega - 1) = 0$, or $2\theta = -\frac{1}{k\Omega - 1}$; which is to say we have $\alpha = \beta = -1$, $\beta = 2(1 - k\Omega)$; consequently

$$\lambda = \Omega k, \quad \text{and} \quad \frac{1-y}{1+y} = \frac{1-x}{1+x} \left[\frac{P-Qx}{P+Qx} \right] = \frac{1-x}{1+x} \left[\frac{-1+2(k\Omega-1)x}{-1-2(k\Omega-1)x} \right],$$

which completes the theory.

[p. 122]

13. Reproducing for this case the general theory, it appears *à priori* that Ω is determined by a quartic equation; in fact, from the original equations eliminating Ω , we have an equation

$$\begin{vmatrix} U & V \\ U' & V' \end{vmatrix} = 0,$$

where U, U', V, V' are quartic functions of α, β ; that is, the ratio $\alpha : \beta$ has four values, and to each of these there corresponds a single value of Ω ; viz. Ω is determined by a quartic equation.

14. Considering next the case $n = 5$, the quintic transformation; the elimination of Ω gives the equations

$$\frac{U}{U'} = \frac{V}{V'} = \frac{W}{W'},$$

where U, U' , &c. are all quadric functions of α, β, γ . We have thence $4 \cdot 4 - 2 \cdot 2 = 12$ sets of values of $\alpha : \beta : \gamma$; viz. considering α, β, γ as coordinates *in piano*, the curves $UV' - U'V = 0$, $UW' - U'W = 0$ are quartic curves intersecting in 16 points; but among these are included the four points $\alpha = 0$, $U = 0$ (in fact, the point $\alpha = 0$, $U' = 0$ four times), which are not points of the curve $VW' - V'W = 0$; there remain therefore $16 - 4 = 12$ intersections, agreeing with the general value $(n+1) \cdot 2^{\frac{1}{2}(n-1)}$. Hence Ω is in the first instance determined by an equation of the order 12; but the proper order being $= 6$, there must be a factor of the order 6 to be rejected. To explain this and to determine the factor, observe that the equations in question are

$$k^2\alpha^2(2\beta\gamma + 2\beta\gamma + \beta^2) - \gamma^2(2\alpha\gamma + 2\alpha\beta + \beta^2) = 0,$$

$$k^2\alpha^2(\alpha + 2\beta) - \gamma^2(\gamma + 2\beta) = 0;$$

at the point $\alpha = 0, \gamma = 0$, the first of these has a double point, the second a triple point; or there are at the point in question 6 intersections; but 4 of these are the points which give the foregoing reduction $16 - 4 = 12$; we have thus the point $\alpha = 0, \gamma = 0$, counting twice among the twelve points. Writing in the two equations $\beta = 0$, the equations become $k^2\alpha^2\gamma^2 - \alpha^2\gamma^2 = 0, k^2\alpha^4 - \gamma^4 = 0$, viz. these will be satisfied if $k^2\alpha^2 - \gamma^2 = 0$, that is, the curves pass through each of the two points ($\beta = 0, \gamma = \pm k\alpha$), and these values satisfy (as in fact they should) the third equation

$$k^2(2\alpha\gamma + 2\alpha\beta + \beta^2)\alpha(\alpha + 2\beta) - \gamma(\gamma + 2\beta)(2\alpha\gamma + 2\beta\gamma + \beta^2) = 0.$$

It is moreover easily shown that the three curves have at each of the points in question a common tangent; viz. taking A, B, C as current coordinates, the tangent at the point (α, β, γ) of the second curve has for its equation

$$A(2\alpha^3 + 3\alpha^2\beta)k^2 + B(k^2\alpha^3 - \gamma^3) + C(-2\gamma^3 - 3\gamma^2\beta) = 0;$$

and for $\beta = 0, \gamma = \pm k\alpha$, this becomes $2kA + B(k + 1) + 2C = 0$, viz. this is the line from the point ($\beta = 0, \gamma = \pm k\alpha$) to the point $(1, -2, 1)$. And similarly for the other two curves we find the same equation for the tangent.

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Hence among the 12 points are included the point ($\gamma = 0, \alpha = 0$) twice, and the points ($\beta = 0, \gamma = \pm k\alpha$) each twice: we have thus a reduction = 6.

15. Writing in the equations $\gamma = 0, \alpha = 0$, the first and third are satisfied identically, and the second becomes $\beta^2 = \Omega\beta^2$, that is, the equations give $\Omega = 1$; writing $\beta = 0$, they become

$$k^2\alpha^2 = \Omega\gamma^2, \quad \alpha\gamma = \Omega\alpha\gamma, \quad \gamma^2 = \Omega k^2\alpha^2,$$

viz. putting herein $\gamma^2 = k^2\alpha^2$, the equations again give $\Omega = 1$; hence the factor of the order 6 is $(\Omega - 1)^6$, and the equation of the twelfth order for the determination of Ω is

$$(\Omega - 1)^6 \{(\Omega, 1)^6\} = 0,$$

where $(\Omega, 1)^6 = 0$ is the Ωk -modular equation above written down.

16. Reverting to the equation

$$\frac{1-y}{1+y} = \frac{1-x}{1+x} \left(\frac{P-Qx}{P+Qx} \right)^2,$$

it is to be observed that for $\alpha = 0, \gamma = 0$, that is, $P = 0$, this becomes simply $y = x$, which is the transformation of the order 1; the corresponding value of the modulus λ is $\lambda = k$, and the equation $\lambda = \Omega^2 k$ then gives $\Omega^2 = 1$, which is replaced by $\Omega - 1 = 0$.

If in the same equation we write $\beta = 0$, that is, $Q = 0$, then (without any use of the equation $\gamma^2 = k^2\alpha^2$) we have $y = x$, the transformation of the order 1; but although this is so, the fundamental equation

$$(P^2 + 2PQx^2 + Q^2x^4)^2 = \Omega^2 k (P^4 + 2PQ + Q^2x^4),$$

which, putting therein $Q = 0$, becomes $(P^2)^2 = \Omega^2 k^{-1} P^4$, that is, $(k^2\alpha + \gamma)^2 = \Omega^2 k^{-1}(\alpha + \gamma x)^2$, is not satisfied by the single relation $\Omega - 1 = 0$, but necessitates the further relation

$$\gamma^2 = k^2\alpha^2.$$

The thing to be observed is that the extraneous factor $(\Omega - 1)^2$, equated to zero, gives for Ω the value $\Omega = 1$ corresponding to the transformation $y = x$ of the order 1.

17. Considering next $n = 7$, the septic transformation; we have here between $\alpha, \beta, \gamma, \delta$ a fourfold relation of the form

$$\begin{vmatrix} U & V & W & Z \\ U' & V' & W' & Z' \end{vmatrix} = 0,$$

where, as before, U, V , &c. are quadric functions, and the number of solutions is here $= 8 \cdot 2^3, = 32$; to each of these corresponds a single value of Ω , or Ω is in the first instance determined by an equation of the order 32. But the order of the modular equation is $= 8$; or representing this by $\{(\Omega, 1)^8\} = 0$, the equation must be

$$(\Omega - 1)^{24} \{(\Omega, 1)^8\} = 0,$$

viz. there must be a special factor of the order 24.

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18. One way of satisfying the equations is to write therein $\alpha = 0, \delta = 0$; the equations thus become

$$k\beta^2 = \Omega\gamma^2, \quad \gamma^2 + 2\beta\gamma = \Omega k(2\beta\gamma + \beta^2);$$

or putting $\beta = \alpha', \gamma = \beta'$,

$$k\alpha'^2 = \Omega\beta'^2, \quad \beta'^2 + 2\alpha'\beta' = \Omega k(\alpha'^2 + \alpha'\beta'),$$

which (with α', β' instead of α, β) are the very equations which belong to the cubic transformation; hence a factor is $\{(\Omega, 1)^4\}$.

Observe that for the values in question $\alpha = 0, \delta = 0, P = \gamma x^2, Q = \beta, = \alpha'$, and therefore $P, Q = (\beta - Qx)^2 = \alpha'^2(\beta' - \beta'x)^2, = \alpha'^2(P' - Q'x)^2$, if $P' = \alpha', Q' = \beta'$,

$$\frac{1-y}{1+y} = \frac{1-x}{1+x} \left(\frac{P' - Q'x}{P' + Q'x} \right)^2,$$

which is the formula for a cubic transformation.

19. The equations may also be satisfied by writing therein $\gamma = k\alpha, \delta = k\beta$; in fact, substituting these values, they become

$$\begin{aligned} k^2\alpha^2 &= \Omega k^2\beta^2, \\ 2k^2\alpha^2 + k(2\alpha\beta + \beta^2) &= \Omega k^2(\alpha^2 + 2\alpha\beta) + 2\Omega k\beta^2, \\ k^2\alpha^2 + 2k(\beta^2 + 2\alpha\beta) &= 2\Omega k^2(\beta^2 + 2\alpha\beta) + \Omega k\beta^2, \\ k(\beta^2 + 2\alpha\beta) &= \Omega k(\alpha^2 + 2\alpha\beta); \end{aligned}$$

the first and last of these are

$$k\alpha^2 = \Omega\beta^2, \quad \beta^2 + 2\alpha\beta = \Omega k(\alpha^2 + 2\alpha\beta),$$

which being satisfied the second and third equations are satisfied identically; and these are the formulae for a cubic transformation; that is, we again have the factor $\{(\Omega, 1)^4\}$.

Observe that for the values in question $\gamma = k\alpha, \delta = k\beta$, we have $P = \alpha(1 + kx^2), Q = \beta(1 + kx^2)$; so that, writing $P' = \alpha, Q' = \beta$, we have for y the value

$$\frac{1-y}{1+y} = \frac{(1-x)}{(1+x)} \frac{(P' - Q'x)^2}{(P' + Q'x)^2},$$

which is the formula for a cubic transformation.

20. It is important to notice that we cannot by writing $\alpha = 0$ or $\delta = 0$ reduce the transformation to a quintic one; in fact, the equation $k^2\alpha^2 = \Omega\delta^2$ shows that if either of these equations is satisfied the other is also satisfied; and we have then the foregoing case $\alpha = 0, \delta = 0$, giving not a quintic but a cubic transformation.

And for the same reason we cannot by writing $\alpha = 0, \beta = 0, \gamma = 0$ or $\beta = 0, \gamma = 0, \delta = 0$ reduce the transformation to the order 1. There is thus no factor $\Omega - 1$.

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21. As regards the non-existence of the factor $\Omega - 1$, I further verify this by writing in the equations $\Omega = 1$; they thus become

$$\begin{aligned} k\alpha^2 &= \delta^2, \\ k(2\alpha\gamma + 2\alpha\beta + \beta^2) &= \gamma^2 + 2\gamma\delta + 2\beta\delta, \\ \gamma^2 + 2\beta\gamma + 2\alpha\delta + 2\beta\delta &= k(2\alpha\gamma + \gamma^2 + 2\alpha\delta + \beta^2), \\ \delta^2 + 2\gamma\delta &= k(\alpha^2 + 2\alpha\beta), \end{aligned}$$

which it is to be shown cannot be satisfied in general, but only for certain values of k . Reducing the last equation, this is $\gamma\delta = k\alpha\beta$, which, combined with the first, gives $\alpha = \pm\sqrt{k}\alpha\gamma = \beta\delta$; and if for convenience we assume $\alpha = 1$, and write also $k = \theta^2$ (that is, $k = \theta^2$), then the values of $\alpha, \beta, \gamma, \delta$ are $\alpha = 1$, $\beta = \gamma\theta^{-1}$, $\gamma = \gamma$, $\delta = \theta$; which values, substituted in the second and the third equations, give two equations in γ, θ ; and from these, eliminating γ , we obtain an equation for the determination of θ , that is, of k . In fact, the second equation gives

$$\theta^2(2\gamma + 2\gamma\theta^{-1} + \gamma^2\theta^{-2}) = \gamma^2 + 2\gamma\theta + 2\gamma,$$

or, dividing by γ and reducing,

$$\gamma(1 - \theta^2) = 2\theta^2(\theta^2 - 1)(\theta^2 - \theta + 1),$$

that is,

$$\gamma(1 + \theta^2) = -2\theta^2(\theta^2 - \theta + 1),$$

or, as this may also be written,

$$(\gamma + \theta^2)(1 + \theta^2) = -\theta^2(\theta - 1)^2,$$

that is,

$$\gamma + \theta^2 = \frac{-\theta^2(\theta - 1)^2}{\theta^2 + 1}.$$

Moreover the third equation gives

$$\gamma^2 + 2\gamma^2\theta^{-1} + 2\theta^2 + 2\gamma = \theta^2(2\gamma + 2\gamma^2\theta + 2\theta + \gamma^2\theta^{-1}),$$

that is,

$$\gamma^2(\theta^4 - 2\theta^3 + 2\theta - 1) - 2(\gamma + \theta^2)\theta^2(\theta^2 - 1) = 0;$$

or dividing by $\theta^2 - 1$, it is

$$\gamma^2(\theta - 1)^2 = 2\theta(\gamma + \theta^2);$$

whence also

$$\gamma = \frac{-2\theta}{(\theta - 1)^2}.$$

Also

$$4\theta^2(\theta^2 - \theta + 1)^2 = \gamma^2(\theta^2 + 1)^2,$$

wherefore

$$2(\theta^2 - \theta + 1)^2 = -\theta(\theta^2 + 1) \quad \text{or} \quad 2(\theta^2 - \theta + 1)^2 + \theta(\theta^2 + 1) = 0,$$

or

$$\theta(\theta^2 + 1)^2 + 2(\theta^2 - \theta + 1)^2 = 0,$$

that is,

$$2\theta^4 - 3\theta^3 + 6\theta^2 - 2\theta + 2 = 0,$$

or finally

$$(2\theta^2 - \theta + 1)(\theta^2 - \theta + 2) = 0.$$

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We have thus $(2\theta + 1)^2 = \theta$, that is, $4\theta^2 + 3\theta + 1 = 0$ or $4k^2 + 3k + 1 = 0$, or else $(\theta^2 + 2)^2 = \theta$, that is, $\theta^4 + 3\theta^2 + 4 = 0$ or $k^2 + 3k + 4 = 0$; viz. the equation in k is

$$(4k^2 + 3k + 1)(k^2 + 3k + 4) = 0,$$

these being in fact the values of k given by the modular equation on putting therein $\Omega = 1$.

The equation of the order 32 thus contains the factor $\{(\Omega, 1)^4\}$ at least twice, and it does not contain either the factor $\Omega - 1$, or the factor $\{(\Omega, 1)^6\}$ belonging to the quintic transformation; it may be conjectured that the factor $\{(\Omega, 1)^4\}$ presents itself six times, and that the form is

$$\{(\Omega, 1)^4\}^6 \{(\Omega, 1)^8\} = 0;$$

but I am not able to verify this, and I do not pursue the discussion further.

22. The foregoing considerations show the grounds of the difficulty of the purely algebraical solution of the problem; the required results, for instance the modular equation, are obtained not in the simple form, but accompanied with special factors of high order. The transcendental theory affords the means of obtaining the results in the proper form without special factors; and I proceed to develop the theory, reproducing the known results as to the modular and multiplier equations, and extending it to the determination of the transformation-coefficients $\alpha, \beta, \dots$

[p. 126] **The Modular Equation. Art. Nos. 23 to 28.**

23. Writing, as usual, $q = e^{-\pi K'/K}$, we have u , a given function of q , viz.

$$u = \sqrt{2} q^{\frac{1}{8}} \frac{1 + q \cdot 1 + q^3 \cdot 1 + q^5 \dots}{1 + q \cdot 1 + q^2 \cdot 1 + q^3 \dots},$$

$$\begin{aligned} v &= \sqrt{2} q^{\frac{1}{8}} (1 - q + 2q^2 - 3q^3 + 4q^4 - 6q^5 + 9q^6 - 14q^7 + \dots) \\ &= \sqrt{2} q^{\frac{1}{8}} f(q) \text{ suppose,} \end{aligned}$$

and this being so, the several values of v and of the other quantities in question are all given in terms of q .

The case chiefly considered is that of n an odd prime; and unless the contrary is stated it is assumed that this is so. We have then $n + 1$ transformations corresponding to the same number $n + 1$ of values of v ; these may be distinguished as $v_0, v_1, v_2, \dots, v_n$; viz. writing α to denote an imaginary n th root of unity, we have

$$\begin{aligned} v_0 &= (-)^{\frac{n^2-1}{8}} \sqrt{2} q^{\frac{n}{8}} f(q^n), \quad v_1 = \sqrt{2} (\alpha q^{\frac{1}{n}})^{\frac{1}{8}} f(\alpha q^{\frac{1}{n}}), \\ v_2 &= \sqrt{2} (\alpha^2 q^{\frac{1}{n}})^{\frac{1}{8}} f(\alpha^2 q^{\frac{1}{n}}), \quad \&c., \\ v_n &= \sqrt{2} q^{\frac{1}{8n}} f(q^{\frac{1}{n}}). \end{aligned}$$

(Observe $(-)^{\frac{n^2-1}{8}} = +1$ for $n = 8p \pm 1$, $= -1$ for $n = 8p \pm 3$.)

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The occurrence of the fractional exponent $\frac{1}{8}$ is, as will appear, a circumstance of great importance; and it will be convenient to introduce the term “octicity,” viz. an expression of the form $q^{\frac{f}{8}} F(q)$ ($f = 0$, or a positive integer not exceeding 7, $F(q)$ a rational function of q) may be said to be of the octicity f .

24. The modular equation is of course

$$(v - v_0)(v - v_1) \dots (v - v_n) = 0;$$

say this is

$$v^{n+1} - Av^n + Bv^{n-1} - \dots = 0,$$

so that $A = \Sigma v_0$, $B = \Sigma v_0 v_1$, &c. In the development of these expressions, the terms having a fractional exponent, with denominator n , would disappear of themselves, as involving symmetrically the several n th roots of unity; and each coefficient would be of the form $q^{\frac{g}{8}} F(q)$, F a rational and integral function of q . It is moreover easy to see that, for the several coefficients $A, B, C, \dots$, g will denote the positive residue (mod. 8) of $n, 2n, 3n, \dots$ respectively.

Hence assuming, as the fact is, that these coefficients are severally rational and integral functions of q , it follows that the form is

$$au^g + bu^{g+8} + cu^{g+16} + \dots,$$

g having the foregoing values for the several coefficients respectively. And it being known that the modular equation is as regards u of the order $= n + 1$, there is a known limit to the number of terms in the several coefficients respectively. We have thus for each coefficient an identity of the form

$$A = au^g + bu^{g+8} + \dots,$$

A and u being each of them given in terms of q , the values of the numerical coefficients $a, b, \dots$ can be determined; and we thus arrive at the modular equation.

25. It is in effect in this manner that the modular equations are calculated in Sohnke's Memoir. Various relations of symmetry in regard to (u, v) and other known properties of the modular equation are made use of in order to reduce the number of the unknown coefficients to a minimum; and (what in practice is obviously an important simplification) instead of the coefficients Σv_0 , $\Sigma v_0 v_1$, &c., it is the sums of powers Σv_0 , Σv_0^2 , &c., which are compared with their expressions in terms of u , in order to the determination of the unknown numerical coefficients $a, b, \dots$. The process is a laborious one (although less so than perhaps might beforehand have been imagined), involving very high numbers; it requires the development up to high powers of q , of the high powers of the before-mentioned function $f(q)$; and Sohnke gives a valuable Table, which I reproduce here; adding to it the three columns which relate to ϕq .

[p. 128] Sohnke's Table (with ϕq columns added)

[Table: Sohnke's Table of the developments in powers of q of $f^N(q)$ for $N = 3, 5, 7, 9, 11, 13$, and of ϕq in three additional columns. The table is here transcribed in landscape format; rows are indexed by ind. of q ($0, 1, 2, \dots, 25$) and each column shows the coefficient of the corresponding power of q in the development of the function indicated. Asterisked footnote: “(Wrongly given by Sohnke as $+34805616$).”]

ind. of q	ϕq	ϕq^2	ϕq^4	$f^3 q$	$f^5 q$	$f^7 q$	$f^9 q$	$f^{11} q$	$f^{13} q$	$f^{15} q$	$f^{17} q$	$f^{19} q$
0	+1	+1	+1	+1	+1	+1	+1	+1	+1	+1	+1	+1
1	+2	0	0	-3	-5	-7	-9	-11	-13	-15	-17	-19
2	+2	+2	0	0	+5	+14	+27	+44	+65	+90	+119	+152
3	0	0	0	+5	+5	+7	-9	-55	-130	-238	-380	-552
4	+2	0	+2	0	-15	-49	-90	-110	-65	+90	+391	+874
5	0	0	0	-7	+5	+50	+126	+253	+428	+621	+782	+836
6	+4	+2	0	0	+15	+24	-78	-264	-560	-940	-1356	-1551
7	0	0	0	+3	-30	-141	-322	-440	-260	+450	+1751	+3704
8	+2	0	0	+15	-10	-49	+108	+803	+2275	+4860	+8755	+13889
9	0	0	0	-30	+60	+343	+1056	+2255	+3978	+5984	+7565	+8038
10	+4	+2	+2	0	+5	+105	+126	-484	-2860	-7800	-15691	-26145
11	0	0	0	+22	-44	-176	-855	-2783	-6760	-13325	-22236	-31958
12	+4	0	0	-30	-25	+147	+1437	+5302	+12870	+24115	+36490	+44574
13	0	0	0	-30	+150	+575	+1485	+2310	+1820	-2730	-15470	-42636
14	+4	0	0	+10	-25	-630	-2925	-7700	-15730	-26280	-36465	-39738
15	0	0	0	+90	-126	-651	-1480	+275	+9100	+30856	+72475	+139194
16	+4	+2	0	-44	+200	+875	+4995	+14157	+31525	+59565	+95485	+126465
17	0	0	0	-77	+275	+1729	+6390	+18887	+45045	+91200	+158015	+233358
18	+4	0	+2	+0	-225	-1407	-7479	-26015	-67925	-149600	-282115	-451250
19	0	0	0	+132	-220	-1995	-9504	-30855	-77220	-159510	-280440	-422256
20	+6	0	0	-110	+250	+2128	+10934	+39226	+109110	+253935	+512435	+918680
21	0	0	0	-115	+675	+2870	+10560	+27357	+50500	+59450	+30450	-67450
22	+4	+2	0	-160	+0	+1638	+10395	+39446	+114550	+277200	+583240	+1097290
23	0	0	0	+198	-825	-3990	-16236	-58300	-176580	-449820	-996300	-1982880
24	+4	0	0	+165	-825	-4500	-18810	-69685	-216905	-589580	-1387845	-2901620
25	0	0	0	-280	+475	+5390	+25025	+88825	+255300	+656535	+1494125	+3060180

* Wrongly given by Sohnke as +34805616.

[p. 130] Series of modular equations (after Sohnke).

26. I give from Sohnke the series of modular equations, adding those for the composite cases $n = 9$ and $n = 15$, as to which see the remarks which follow the Table.

[Table: modular equation for $n = 3$. The grid has rows $u^4, u^3, u^2, u, 1$ and columns $v^4, v^3, v^2, v, 1$; the non-zero entries are]

$$\begin{aligned} u^4 : & -1 \text{ at column } 1, \\ u^3 : & +2 \text{ at column } v^2, \\ u^2 : & 0, \\ u^1 : & -2 \text{ at column } v^2, \\ 1 : & +1 \text{ at column } v^4. \end{aligned}$$

The column-sum reads

$$1 \cdot v^4 + 2 \cdot v^2 \cdot u^3 + 0 - 2 \cdot v^2 u - 1 \cdot u^4 = (v + 1)^3(v - 1).$$

[Table: modular equation for $n = 5$. Rows $u^6, u^5, u^4, u^3, u^2, u, 1$, columns $v^6, v^5, v^4, v^3, v^2, v, 1$; the non-zero entries are]

$$\begin{aligned} u^6 : & -1 \text{ at column } 1, \\ u^5 : & +4 \text{ at column } v^2, \\ u^4 : & -5 \text{ at column } v^4, \\ u^3 : & +5 \text{ at column } v^3, \\ u^2 : & -4 \text{ at column } v^5, \\ u : & +1 \text{ at column } v^6. \end{aligned}$$

The full equation reads

$$1 + 4 + 5 + 0 - 5 - 4 - 1 = (v + 1)^5(v - 1).$$

[Table: modular equation for $n = 7$. Rows $u^8, \dots, u, 1$, columns $v^8, \dots, v, 1$; the non-zero entries form a skew-diagonal pattern with binomial-like values:]

$$\begin{aligned} u^8 : & +1 \text{ at column } 1, \\ u^7 : & -8 \text{ at column } v^2, \\ u^6 : & +28 \text{ at column } v^4, \\ u^5 : & -56 \text{ at column } v^5, \\ u^4 : & +70 \text{ at column } v^4, \\ u^3 : & -56 \text{ at column } v^5, \\ u^2 : & +28 \text{ at column } v^6, \\ u : & -8 \text{ at column } v^7, \\ 1 : & +1 \text{ at column } v^8. \end{aligned}$$

The complete column-sum reads

$$+1 - 8 + 28 - 56 + 70 - 56 + 28 - 8 + 1 = (v - 1)^8.$$

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[Table: modular equation for $n = 9$. Rows $u^{12}, u^{11}, \dots, u, 1$ and columns $v^{12}, v^{11}, \dots, v, 1$. The diagonal pattern of non-zero entries is]

$$\begin{aligned}
 u^{12} &: +1 \text{ at col } 1, \\
 u^{11} &: -8 \text{ at col } v^3, \\
 u^{10} &: +16 \text{ at col } v^6, \\
 u^9 &: -16 \text{ at col } v^8, \quad +10 \text{ at col } v^4, \\
 u^8 &: +8 \text{ at col } v^7, \quad -16 \text{ at col } v^5, \\
 u^7 &: -16 \text{ at col } v^6, \quad +15 \text{ at col } v^4, \\
 u^6 &: +48 \text{ at col } v^5, \quad -84 \text{ at col } v^3, \\
 u^5 &: -84 \text{ at col } v^4, \quad +48 \text{ at col } v^6, \\
 u^4 &: +48 \text{ at col } v^3, \quad +15 \text{ at col } v^5, \\
 u^3 &: +15 \text{ at col } v^4, \quad -24 \text{ at col } v^5, \\
 u^2 &: -24 \text{ at col } v^5, \quad +10 \text{ at col } v^3, \quad -16 \text{ at col } v^4, \\
 u &: +10 \text{ at col } v^4, \quad +16 \text{ at col } v^2, \quad -16 \text{ at col } v^3, \\
 1 &: +8 \text{ at col } v^3, \quad -16 \text{ at col } v^4, \quad +1 \text{ at col } v.
 \end{aligned}$$

The full column-sum, when read across $v^{12}, v^{11}, \dots, v, 1$, is

$$1 - 8 + 26 - 40 + 15 + 48 - 84 + 48 + 15 - 40 + 26 - 8 + 1 = (v - 1)^{10}(v + 1)^2.$$

[Table: modular equation for $n = 11$. Rows $u^{12}, u^{11}, \dots, u, 1$ and columns $v^{12}, v^{11}, \dots, v, 1$. Non-zero entries:]

$$\begin{aligned}
 u^{12} &: -1 \text{ at col } 1, \\
 u^{11} &: +32 \text{ at col } v^2, \\
 u^{10} &: -44 \text{ at col } v^4, \\
 u^9 &: +88 \text{ at col } v^3, \quad +22 \text{ at col } v^7, \\
 u^8 &: -165 \text{ at col } v^5, \\
 u^7 &: +132 \text{ at col } v^4, \\
 u^6 &: +44 \text{ at col } v^5, \quad -44 \text{ at col } v^8, \\
 u^5 &: -132 \text{ at col } v^6, \\
 u^4 &: +165 \text{ at col } v^7, \\
 u^3 &: -22 \text{ at col } v^8, \quad -88 \text{ at col } v^9, \\
 u^2 &: +44 \text{ at col } v^9, \\
 u &: +22 \text{ at col } v^{10}, \quad -32 \text{ at col } v^{11}, \\
 1 &: +1 \text{ at col } v^{12}.
 \end{aligned}$$

The full row of column-sums reads

$$1 + 10 + 44 + 110 + 165 + 132 + 0 - 132 - 165 - 110 - 44 - 10 - 1 = (v + 1)^{11}(v - 1).$$

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[Table: modular equation for $n = 13$. Rows $u^{14}, u^{13}, \dots, u, 1$ and columns $v^{14}, v^{13}, \dots, v, 1$. Non-zero entries are arranged in the skew-diagonal pattern characteristic of $n \equiv 5 \pmod{8}$.]

$$\begin{aligned}
 u^{14} &: -1 \text{ at col } 1, \\
 u^{13} &: +64 \text{ at col } v^2, \quad -52 \text{ at col } v^8, \\
 u^{12} &: -65 \text{ at col } v^4, \\
 u^{11} &: +208 \text{ at col } v^6, \\
 u^{10} &: +208 \text{ at col } v^5, \quad -429 \text{ at col } v^7, \\
 u^9 &: +520 \text{ at col } v^6, \quad +52 \text{ at col } v^9, \\
 u^8 &: -429 \text{ at col } v^7, \\
 u^7 &: -208 \text{ at col } v^7, \\
 u^6 &: +429 \text{ at col } v^8, \\
 u^5 &: -52 \text{ at col } v^{11}, \quad -520 \text{ at col } v^9, \\
 u^4 &: +429 \text{ at col } v^{10}, \\
 u^3 &: -208 \text{ at col } v^{11}, \\
 u^2 &: +65 \text{ at col } v^{12}, \\
 u &: +52 \text{ at col } v^{13}, \quad -64 \text{ at col } v^{13}, \\
 1 &: +1 \text{ at col } v^{14}.
 \end{aligned}$$

The complete column-sum reads

$$1 + 12 + 65 + 208 + 429 + 572 + 429 + 0 - 429 - 572 - 429 - 208 - 65 - 12 - 1 = (v + 1)^{13}(v - 1).$$

[p. 133] Modular equation for $n = 17$.

[Table: a large rotated table (sidewaystable in the original) giving the modular equation for $n = 17$. Rows are powers $u^{18}, u^{17}, \dots, u, 1$ and columns are powers $v^{18}, v^{17}, \dots, v, 1$. The entries follow the skew-symmetric pattern of $n \equiv 1 \pmod{8}$ (perfect symmetry, no sign reversal); the non-zero entries appearing in the printed table, read along the principal diagonals, are coefficients such as $+1, -8, +28, +66, +220, \dots$ on the dexter (descending) diagonal and their reflections on the sinister (rising) diagonal. The column-sum reduces to $(v - 1)^{18}$.]

[p. 134] Modular equation for $n = 19$.

[Table: rotated landscape table for the modular equation of $n = 19$. Rows are $u^{20}, u^{19}, \dots, u, 1$ and columns are $v^{20}, v^{19}, \dots, v, 1$. Owing to the case $n \equiv 3 \pmod{8}$, the table is not perfectly symmetric but has alternating sign reversals on the sinister diagonal. The non-zero diagonal coefficients begin $-1, +114, -114, 0, -2584, -6859, -2584, \dots$ as referred to in Art. 26. The full column-sum reduces to $(v+1)^{19}(v-1)$.]

[p. 135] Modular equation for $n = 21$ and remarks.

[Table: rotated landscape table for the modular equation of $n = 21$. Rows are $u^{22}, u^{21}, \dots, u, 1$ and columns are $v^{22}, v^{21}, \dots, v, 1$. The pattern is symmetric with sign reversals on alternate lines parallel to the sinister diagonal. The column-sum reduces to $(v + 1)^{21}(v - 1)$.]

[p. 136] Remarks on the Tables.

Various remarks arise on the Tables. Attending first to the cases n a prime number; the only terms of the order $n + 1$ in v or u are $v^{n+1} \mp u^{n+1}$, viz. $n = 3$ or $5 \pmod{8}$ the sign is $-$, but $n = 1$ or $7 \pmod{8}$ the sign is $+$. And there is in every case a pair of terms $v^n u^a$ and $v^a u^n$, having coefficients equal in absolute magnitude, but of opposite signs, or of the same sign, in the two cases respectively.

Each Table is symmetrical in regard to its two diagonals respectively, so that every non-diagonal coefficient occurs (with or without reversal of sign) 4 times; viz. in the case $n = 1$ or $7 \pmod{8}$ this is a perfect symmetry, without reversal of sign; but in the case $n = 3$ or $5 \pmod{8}$ it is, as regards the lines parallel to either diagonal, and in regard to the other diagonal, alternately a perfect symmetry without reversal of sign and a skew symmetry with reversal. Thus in the case $n = 19$, the lines parallel to the dexter diagonal are -1 (symmetrical), $+114, -114$ (skew), $0, -2584, -6859, -2584, \dots$ (symmetrical), and so on. The same relation of symmetry is seen in the composite cases $n = 9$ and $n = 15$, both belonging to $n = 1$ or $7 \pmod{8}$.

If, as before, n is prime, then putting in the modular equation $u = 1$, the equation in the case $n = 1$ or $7 \pmod{8}$ becomes $(v - 1)^{n+1} = 0$, but in the case $n = 3$ or $5 \pmod{8}$ it becomes $(v + 1)^n(v - 1) = 0$.

27. In the case n a composite number not containing any square factor, then dividing n in every possible way into two factors $n = ab$ (including the divisions $n \cdot 1$ and $1 \cdot n$), and denoting by β an imaginary b th root of unity, a value of v is

$$\pm \sqrt{2} (\beta q^{\frac{a}{b}})^{\frac{1}{8}} f\left(\beta q^{\frac{a}{b}}\right);$$

so that the whole number of roots (or order of the modular equation) is $= \nu$, if ν be the sum of the divisors of n . Thus $n = 15$, the values are

$$\sqrt{2} q^{\frac{15}{8}} f(q^{15}), \quad -\sqrt{2} q^{\frac{5}{8}} f(q^5), \quad -\sqrt{2} q^{\frac{3}{8}} f(q^3), \quad \sqrt{2} q^{\frac{1}{8}} f(q),$$

$$1, \quad 3, \quad 5, \quad 15 \text{ roots;}$$

and the order of the modular equation is $= 24$. The modular equation might thus be obtained as for a prime number; but it is easier to decompose n into its prime factors, and consider the transformation as compounded of transformations of these prime orders. Thus $n = 15$, the transformation is compounded of a cubic and a quintic one. If the v of the cubic transformation be denoted by θ , then we have

$$\theta^4 + 2\theta^3 u^3 - 2\theta u - u^4 = 0;$$

and to each of the four values of θ corresponds the six values of v belonging to the quintic transformation given by

...

The equation in v is thus

...

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where $\theta_1, \theta_2, \theta_3, \theta_4$ are the roots of the equation in θ , viz. we have

$$\theta^4 + 2\theta^3 u^3 - 2\theta u - u^4 = (\theta - \theta_1)(\theta - \theta_2)(\theta - \theta_3)(\theta - \theta_4);$$

and it was in this way that the equation for the case $n = 15$ was calculated. Observe that writing $u = 1$, we have $(\theta + 1)^3(\theta - 1) = 0$, or say $\theta_1 = \theta_2 = \theta_3 = -1$, $\theta_4 = +1$. The equation in v thus becomes $\{(v - 1)^3(v + 1)\}^3(v - 1)^3(v + 1) = 0$, that is, $(v - 1)^{12}(v + 1)^{12} = 0$.

28. The case where n has a square factor is a little different; thus $n = 9$, the values are

$$1, \quad 3, \quad 9 \dots \text{roots,}$$

but here ω being an imaginary cube root of unity, the second term denotes the three values,

$$\sqrt{2} q^{\frac{1}{8}} f(q), \quad \sqrt{2} (\omega q)^{\frac{1}{8}} f(\omega q), \quad \sqrt{2} (\omega^2 q)^{\frac{1}{8}} f(\omega^2 q),$$

the first of which is $= u$, and is to be rejected; there remain $1 + 2 + 9 = 12$ roots, or the equation is of the order 12.

Considering the equation as compounded of two cubic transformations, if the value of v for the first of these be θ , then we have

$$\theta^4 + 2\theta^3 u^3 - 2\theta u - u^4 = 0;$$

and to the four values of θ correspond severally the four values of v given by the equation

$$v^4 + 2v^3 \theta^3 - 2v\theta - \theta^4 = 0.$$

One of these values is however $v = -u$, since the u^4 -equation is satisfied on writing therein $v = -u$; hence, writing

$$\theta^4 + 2\theta^3 u^3 - 2\theta u - u^4 = (\theta - \theta_1)(\theta - \theta_2)(\theta - \theta_3)(\theta - \theta_4),$$

we have an equation

$$(v^4 + 2v^3 \theta_1^3 - 2v\theta_1 - \theta_1^4)(v^4 + \dots \theta_2^3 \dots \theta_2^4) \dots (v^4 + \dots \theta_4^3 \dots \theta_4^4) = 0,$$

which contains the factor $(v + u)^4$ and, divested hereof, gives the required modular equation of the order 12; it was in fact obtained in this manner.

Observe that writing $u = 1$, we have $(\theta + 1)^3(\theta - 1) = 0$, or say $\theta_1 = \theta_2 = \theta_3 = -1$, $\theta_4 = 1$; the modular equation then becomes

$$\{(v - 1)^3(v + 1)\}^3 (v + 1)^3 (v - 1)^3 (v + 1)^3 = 0,$$

that is,

$$(v - 1)^9 (v + 1)^9 = 0.$$

[p. 138] The Multiplier Equation. Art. No. 29.

29. The theory is in many respects analogous to that of the modular equation. To each value of v there corresponds a single value of M ; hence M , or what is the same thing $\frac{1}{M}$, is determined by an equation of the same order as v , viz. n being prime, the order is $= n + 1$. The last term of the equation is constant, and the other coefficients are rational and integral functions of u^4 , of a degree not exceeding $\frac{1}{2}(n - 1)$; and not only so, but they are, $n \equiv 1 \pmod{4}$, rational and integral functions of $u^4(1 - u^4)$, and $n \equiv 3 \pmod{4}$, alternately of this form and of the same form multiplied by the factor $(1 - 2u^4)$.

The values are in fact given as transcendental functions of q ; viz. denoting by $M_0, M_1, M_2, \dots, M_n$ the values corresponding to $v_0, v_1, v_2, \dots, v_n$ respectively, and writing

$$\begin{aligned}\phi(q) &= \frac{(1+q)(1+q^2)(1+q^3)\cdots(1-q^2)(1-q^4)\cdots}{(1-q)(1-q^3)(1-q^5)\cdots(1+q^2)(1+q^4)\cdots}, \\ &= 1 + 2q + 2q^4 + 2q^9 + 2q^{16} + \cdots,\end{aligned}$$

then we have

$$\begin{aligned}(-)^{\frac{n-1}{2}} M_0^{\frac{1}{2}} &= \frac{\phi(q)}{\phi(q^n)}, \\ M_1^{\frac{1}{2}} &= \frac{\phi(q)}{\phi(\alpha q^{\frac{1}{n}})}, \quad (\alpha \text{ an imaginary } n\text{th root of unity}), \\ M_n^{\frac{1}{2}} &= \frac{\phi(q)}{\phi(q^{\frac{1}{n}})}.\end{aligned}$$

Hence, the form of the equation being known, the values of the numerical coefficients may be calculated; and it was in this way that Joubert obtained the following results. I have in some cases changed the sign of Joubert's multiplier, so that in every case the value corresponding to $u = 0$ shall be $M = 1$.

The equations are:

$$\begin{aligned}n = 3, \quad \frac{1}{M^4} - 3 &= 0; \quad u = 0, \text{ this is } \left(\frac{1}{M} - 1\right)^3 \left(\frac{1}{M} + 3\right) = 0; \\ u = 1, \text{ it is } \left(\frac{1}{M} + 1\right)^3 \left(\frac{1}{M} - 3\right) &= 0.\end{aligned}$$

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$$\begin{aligned}n = 5, \quad \frac{1}{M^6}, \\ + \frac{1}{M^4} \cdot (-10), \\ + \frac{1}{M^3} \cdot (+35), \\ + \frac{1}{M^2} \cdot (-60), \\ + \frac{1}{M} \cdot (+55), \\ \frac{1}{M} \cdot [-26 + 256u^4(1 - u^4)] + 5 &= 0; \\ u = 0 \text{ or } 1, \text{ this is } \left(\frac{1}{M} - 1\right)^5 \left(\frac{1}{M} - 5\right) &= 0. \\ n = 7, \quad \frac{1}{M^8}, \\ + \frac{1}{M^7} \cdot 0,\end{aligned}$$

$$\begin{aligned}
 & + \frac{1}{M^6} \cdot (-28), \\
 & + \frac{1}{M^5} \cdot [+112(1 - 2u^4)], \\
 & + \frac{1}{M^4} \cdot (-210), \\
 & + \frac{1}{M^3} \cdot [+224(1 - 2u^4)], \\
 & + \frac{1}{M^2} \cdot [-140 - 21 \cdot 256 u^4(1 - u^4)], \\
 & + \frac{1}{M} \cdot [48 + 2048 u^4(1 - u^4)] (1 - 2u^4) + 7 = 0; \\
 & u = 0, \text{ this is } \left(\frac{1}{M} - 1 \right)^7 \left(\frac{1}{M} - 7 \right) = 0; \\
 & u = 1, \text{ it is } \left(\frac{1}{M} + 1 \right)^7 \left(\frac{1}{M} + 7 \right) = 0. \\
 & n = 11, \quad \frac{1}{M^{12}}, \\
 & + \frac{1}{M^{11}} \cdot 0, \\
 & + \frac{1}{M^{10}} \cdot (-66), \\
 & + \frac{1}{M^9} \cdot [+440(1 - 2u^4)],
 \end{aligned}$$

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$$\begin{aligned}
 & + \frac{1}{M^8} \cdot (-1485), \\
 & + \frac{1}{M^7} \cdot [+3168(1 - 2u^4)], \\
 & + \frac{1}{M^6} \cdot [-4620 - 3 \cdot 11^2 \cdot 256 u^4(1 - u^4)], \\
 & + \frac{1}{M^5} \cdot [+4752 + 11 \cdot 4096 u^4(1 - u^4)] (1 - 2u^4), \\
 & + \frac{1}{M^4} \cdot [-3465 - 3 \cdot 7 \cdot 11 \cdot 512 u^4(1 - u^4)], \\
 & + \frac{1}{M^3} \cdot [+1760 - 11 \cdot 83 \cdot 2048 u^4(1 - u^4)] (1 - 2u^4), \\
 & + \frac{1}{M^2} \cdot [-594 - 9 \cdot 11 \cdot 37 \cdot 256 u^4(1 - u^4) - 3 \cdot 11 \cdot 131072 [u^4(1 - u^4)]^2], \\
 & + \frac{1}{M} \cdot [120 + 15 \cdot 4096 u^4(1 - u^4) - 524288 (u^4(1 - u^4))^2] (1 - 2u^4) \\
 & - 11 = 0. \\
 & u = 1, \text{ it is } \left(\frac{1}{M} + 1 \right)^{11} \left(\frac{1}{M} - 11 \right) = 0.
 \end{aligned}$$

[p. 140] The Multiplier as a rational function of u, v . Art. Nos. 30 to 36.

30. The multiplier M , as having a single value corresponding to each value of v , is necessarily a rational function of u, v ; and such an expression of M can, as remarked by Königsberger, be deduced from the multiplier equation by means of Jacobi's theorem,

$$M^n = \frac{1}{n} \frac{\lambda(1-\lambda^2)}{k(1-k^2)} \frac{dk}{d\lambda},$$

viz. substituting for k, λ their values u^4, v^4 , and observing that if the modular equation be $F(u, v) = 0$ so that the value of $\frac{dv}{du}$ is $-F'(v) \div F'(u)$, this is

$$M^n = \frac{1}{n} \frac{(1-v^4)vF'v}{(1-u^4)uF'u},$$

and then in the multiplier equation separating the terms which contain the odd and even powers, and writing it in the form $\Phi(M^2) + M\Psi(M^2) = 0$, this equation, substituting therein for M^2 its value, gives the value of M rationally.

The rational expression of M in terms of u, v is of course indeterminate, since its form may be modified in any manner by means of the equation $F(u, v) = 0$; and in the expression obtained as above, the orders of the numerator and the denominator are far too high. A different form may be obtained as follows: for greater convenience I seek for the value not of M but of $\frac{1}{M}$.

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31. Denoting, as above, by $M_0, M_1, \dots, M_n$ the values which correspond to $v_0, v_1, \dots, v_n$ respectively, and writing $S_{\frac{1}{M}} = \frac{1}{M_0} + \frac{1}{M_1} + \dots + \frac{1}{M_n}$, &c., we have $S_{\frac{1}{M}}, S_{\frac{v}{M}}$, &c., all of them expressible as determinate functions of u ; and we have moreover the theorem that each of these is a rational and integral function of u : we have thus the series of equations

$$S_{\frac{1}{M}} = A, \quad S_{\frac{v}{M}} = B, \quad \dots, \quad S_{\frac{v^n}{M}} = H,$$

where $A, B, \dots, H$ are rational and integral functions of u . These give linearly the different values of $\frac{1}{M}$; in fact, we have

$$(v_0 - v_1) \cdots (v_0 - v_n) \frac{1}{M_0} = H - G \Sigma v_1 + F \Sigma v_1 v_2 - \dots \pm A v_1 v_2 \cdots v_n,$$

where $\Sigma v_1, \Sigma v_1 v_2$, &c. denote the combinations formed with the roots $v_1, v_2, \dots, v_n$ (these can be expressed in terms of the single root v_0); and we have also $(v_0 - v_1) \cdots (v_0 - v_n) = F'(v_0)$ the resulting equation is consequently

$$F'v_0 \frac{1}{M_0} = R(u, v_0),$$

a determinate rational and integral function of (u, v_0) ; but as the same formula exists for each root of the modular equation, we may herein write M, v in place of M_0, v_0 ; and the formula thus is

$$F'v \cdot \frac{1}{M} = R(u, v),$$

viz. we thus obtain the required value of $\frac{1}{M}$ as a rational function, the denominator being the determinate function $F'v$, and the numerator being, as is easy to see, a determinate function of the order n as regards v .

32. The method is applicable when M is only known by its expression in terms of q ; but if we know for M an expression in terms of v, u , then the method transforms this into a standard form as above. By way of illustration I will consider the case $n = 3$, where the modular equation is

$$v^4 + 2v^3u^3 - 2vu - u^4 = 0,$$

$$v = u + \frac{1 - 2u^4}{\dots},$$

and where a known expression of M is $\frac{1}{\dots}$. Here writing $S_{-1}, S_0 (= 4), S_1$, &c. to denote the sum of the powers $-1, 0, 1$, &c. of the roots of the equation, we have

$$S_{\frac{1}{M}} = 0 = S_3 + 2u^3 S_{-1}, \quad \text{as appears from the values presently given,}$$

$$S_{\frac{v}{M}} = 0 = S_4 + 2u^3 S_0,$$

$$S_{\frac{v^2}{M}} = S_5 + 2u^3 S_1 = 6u;$$

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and observing that v_0 , being ultimately replaced by v , we have

$$\Sigma v_1 = \Sigma v_0 - v, \quad \Sigma v_1 v_2 = \Sigma v_0 v_1 - v \Sigma v_0 + v^2, \quad v_1 v_2 v_3 = \Sigma v_0 v_1 v_2 - v \Sigma v_0 v_1 + v^2 \Sigma v_0 - v^3,$$

that is,

$$\Sigma v_1 = -2u^3 - v, \quad \Sigma v_1 v_2 = -2u^3 v - v^2, \quad v_1 v_2 v_3 = 2u - 2u^3 v^2 - v^3,$$

we have

$$\begin{aligned} F'v \cdot \frac{1}{M} &= (S_3 + 2u^3 S_0) \dots + (2u^3 + v)(S_2 + 2u^3 S_1) \\ &\quad + (2u^3 v + v^2)(S_1 + 2u^3 S_0) \\ &\quad + (-2u + 2u^3 v^2 + v^3)(S_0 + 2u^3 S_{-1}), \end{aligned}$$

viz. this is

...

But we have

$$S_{-1} = -3, \quad S_0 = 4, \quad S_1 = -2u^3, \quad S_2 = u^2, \quad S_3 = 6u - 8u^3, \dots$$

and the equation thus is

$$(2v^3 + 3v^2 u^3 - u) \cdot \frac{1}{M} = 3(v^2 u^2 + 2u^3 v + 1);$$

to verify which observe that, substituting herein for $\frac{1}{M}$ its value $1 + \frac{2u^3}{v}$, the equation becomes

$$Sv^3 u^3 + (2v^2 + 0)(v^2 u^3)Svu \cdot (v^2 u^2 + 2u^3 v + 1);$$

that is,

$$2v^4 + 4u^3 v^3 - 4vu - 2u^4 = 0,$$

as it should do.

33. Any expression whatever of M in terms of u, v is in fact one of a system of four expressions; viz. we may simultaneously change

that is, signs are

u	v	$\frac{1}{M}$	$n = 1$	$n = 3$	$n = 5, n = 7 \pmod{8}$
u	v	$\frac{1}{M}$	$- + + +$	$+ - + -$	$+ - - +$
into v	$(-)^{\frac{n^2-1}{8}} u$	$(-)^{\frac{n-1}{2}} \frac{n^2}{M^n}$	$+ + + +$	$+ + + -$	$+ + + -$
or $\frac{1}{u}$	$\frac{1}{v}$	$\frac{v^n}{u^n M}$	$+ + + +$	$+ + + -$	$+ + + -$
or $\frac{1}{v}$	$(-)^{\frac{n^2-1}{8}} \frac{1}{u}$	$(-)^{\frac{n-1}{2}} \frac{u^n}{v^n M^n}$	$+ + +$	$+ - +$	$+ - +$